BanglaAV: Audio-Visual Deepfake Localization

Week 4 - Method Extension and Controlled Evidence

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Research checkpoint: 11 November 2026

Abstract - We implemented **Local-Window Consensus Alignment (LWCA)** to reduce the influence of locally manipulated audio/visual activity on the clip-wide synchronization estimate. In a newly generated 1,800-sequence synthetic holdout, the selected 16-bin method reduced mean absolute lag error from 78.0 to 25.1 ms (audio edits), 43.1 to 13.1 ms (visual edits), and 110.0 to 26.0 ms (dual edits). All eight integration tests passed. **These are controlled feature-level results, not LAV-DF or Bengali deepfake-detection results.**

Index Terms - temporal localization, audio-visual synchronization, robust lag consensus, ablation, BanglaAV.

I. MOTIVATION AND RESEARCH HYPOTHESIS

In Update 3, our delay-aware pipeline estimated a single offset from the entire clip. This introduces a potential failure: a strong local edit can distort the global audio-energy/mouth-motion correlation, even if most of the recording is unedited. Our testable hypothesis is that locally estimated delays, combined by consensus, better recover the underlying recording offset under partial manipulation. This is a new *project-level method extension*, not a claim of first discovery in the literature.

II. PROPOSED METHOD: LWCA

Both streams are mapped to 80-ms bins, as in Update 3. We compare the absolute temporal change of audio energy with lower-face motion. Overlapping windows of 16 bins (1.28 s), advanced by 8 bins (0.64 s), each search offsets from -5 to +5 bins (±400 ms). The maximum Pearson correlation votes for a lag only when it exceeds 0.24. The modal lag is accepted if its voting support is at least 50% and it has adequate advantage over zero offset; otherwise we return zero.

The resulting aligned and unaligned features enter the same 48-hidden-unit head architecture as the previous delay-aware method. We added **robust_delay_aware** as a seventh model with separate audio/visual heads; existing baselines stay intact. Cross-modal false attribution (e.g., an audio alarm on a visual-only edit) is now explicitly counted.

III. EXPERIMENT DESIGN

We extracted the *original* global alignment function from the team's submitted Update 3 notebook; the comparator was not reimplemented. Each protocol generates 3 random seeds × 5 conditions × 120 samples = 1,800 independent synthetic feature sequences. Conditions are aligned genuine, clean shifted, audio edit, visual edit, and dual edit. Ground-truth global offset is known by construction.

We used 16-, 32-, and 48-bin consensus windows as an ablation. After development experiments (seeds 17, 29, 43), the more variable holdout used new seeds 101, 211, 307 and changes in signal length, pulse rate, background noise, edit length, and offset range. No examples were reused across protocols. A fixed-seed, 1,000-resample paired bootstrap summarizes holdout lag-error differences.

IV. MEASURED HOLDOUT RESULTS

Table I compares lag-estimation error on the three edited conditions (360 sequences each). Lower is better. Exact lag-match accuracy improved from 78.1% to 91.1% for audio edits, 86.1% to 94.7% for visual edits, and 61.4% to 90.3% for dual edits.

TABLE I. Synthetic holdout MAE in ms; delta = LWCA - global (negative favors LWCA).

Edit type	Global MAE	LWCA MAE	Delta [95% CI]
Audio	78.0	25.1	-52.9 [-69.8, -36.9]
Visual	43.1	13.1	-30.0 [-40.2, -19.8]
Dual	110.0	26.0	-84.0 [-103.6, -66.2]

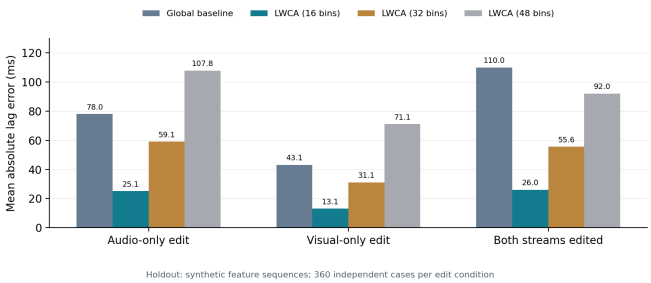


Fig. 1. Window-length ablation on the synthetic holdout (360 examples per edit type).

V. CHECKS AND INTERPRETATION

For clean genuine and clean shifted cases, both the global estimator and LWCA had zero lag error in this simulator. That ceiling result reflects relatively simple synthetic patterns and cannot establish clean real-video reliability. The 16-bin window beat the tested 32- and 48-bin alternatives under these localized corruptions. This supports the proposed mechanism, but it may not generalize to speech, occlusion, off-center faces, or conversational mouth movements.

A functional smoke run of the patched notebook passed 8/8 tests, including training all 10 heads across seven model variants on small synthetic arrays, producing finite predictions, model-specific lag metadata, false-modality alarms, and paired comparison rows. This is not a real-media training run.

VI. LIMITATIONS AND NEXT RUN

Uncompleted real-media validation: The permitted LAV-DF media and existing trained heads were unavailable here. No new LAV-DF AP50, F1, genuine-video false alarm rate, or Bengali metric has been measured. A synthetic lag-error improvement is *not* yet a deepfake-localization gain.

Next, run the augmented notebook on the existing 320/96 train/development protocol, compare raw/delay/LWCA fusion with multiple training seeds, and evaluate AP50, edit recall, genuine-video alarm rate, and wrong-modality attribution at matched recall. Stress with ±240-ms feature shifts on the same temporal support and shifted audio labels. Select thresholds only on development data; open the reserved test set once. Bangla speaker/source-disjoint collection with participant consent remains necessary.

VII. REPRODUCIBILITY AND PROVENANCE

The companion ZIP includes the unchanged Update 3 source notebook, patched Update 4 notebook, readable diff, LWCA algorithm, synthetic generator, notebook patcher, training integration smoke runner, aggregate and per-example CSVs, seed/protocol JSON, plot code, and this report. Original source: NSU-CSE499-Research/CSE499-Senior-Design, commit c96b64ea69ac9ce28d484dfbfacf7ce02da5bdd0. No protected video data, participant recordings or model checkpoints are redistributed.

REFERENCES

[1] BanglaAV CSE499 team, *Update 3: Implementation and LAV-DF Pilot Results*, NSU, November 2026.

[2] NSU-CSE499-Research, *CSE499-Senior-Design*, Update-03/cse499b-update-3.ipynb, GitHub commit c96b64ea, accessed November 2026.